PERSONAL INFORMATION

Name: Guowen Xu (IEEE/CCF Senior Member)
Date of Birth: June 5, 1991
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Address: University of Electronic Science and Technology of China, P.R. China

EDUCATION BACKGROUND

Ph.D. of Cyberspace Security **2015/09–2020/12**
School of Computer Science and Engineering
University of Electronic Science and Technology of China (UESTC)

Visiting Ph.D. in Cyberspace Security **2019/08–2020/08**
School of Information of Systems
Singapore Management University

Bachelor of Information and Computing Science **2010/09–2014/06**
School of Mathematical and Physical Science
Anhui Jianzhu University (AHJZU)

PROFESSIONAL EXPERIENCE

Research Fellow **2021/03–2023/05**
School of Computer Science and Engineering
Nanyang Technological University, Singapore

Research Fellow **2023/05–2024/08**
Department of Computer Science
City University of Hong Kong

Professor **2024/08–present**
School of Computer Science and Engineering
University of Electronic Science and Technology of China

RESEARCH INTERESTS

AI Security and Privacy, Data Security

AWARDS AND HONORS

- 2026 Journal of Cybersecurity Outstanding Paper Award
- 2026 IEEE-HKN Frank and Shelley Chang Outstanding Young Scholar Award
- 2025 IEEE Transactions on Network and Service Management Exemplary Associate Editor Award
- 2025 IEEE 31st International Conference on Parallel and Distributed Systems Best Presentation in Session Award
- 2025 IEEE Signal Processing Society Best Paper Award
- 2025 R10 Young Professionals Outstanding Volunteer in Academic Award
- 2025 IEEE Systems, Man, and Cybernetics Society's Technical Committee on Homeland Security Young Researcher Award
- 2024 IEEE Computer Society Computing's Top 30 Early Career Professionals
- 2024 IEEE Computer Society Early Career Speaker
- 2024 Cybersecurity journal (Springer) Outstanding Youth Editor Award
- 2023 IEEE 9th Intl Conference on Big Data Security on Cloud Best paper Award
- 2021 Wu Wenjun First Prize of Artificial Intelligence Science and Technology Progress
- 2020 IEEE 26th International Conference on Parallel and Distributed Systems Best Paper Award

SELECTED PUBLICATIONS

CCF A papers: 82; Google Citations: 8083; * Denote the corresponding author

[1] **[TDSC 2026]** Xinyuan Qian, Hongwei Li, **Guowen Xu***, Haoyong Wang, Hangcheng Cao, Cong Wu, Tianwei Zhang, Yuguang Fang. Perspicio: Safeguarding Against Non-consensual Photo Sharing over Social Networks. *IEEE Transactions on Dependable and Secure Computing*, 2026. (**CCF A**)

[2] **[TDSC 2026]** Cong Wu, Jing Chen, Jian Shen, **Guowen Xu***, Yueming Wu, Haijun Wang, Hongwei Li, Yang Liu, Tao Xiang. Catching Scam Tokens with Temporal Graph Learning in Decentralized Finance. *IEEE Transactions on Dependable and Secure Computing*, 2026. (**CCF A**)

[3] **[TDSC 2026]** Shuai Yuan, **Guowen Xu***, Hongwei Li, Rui Zhang, Xinyuan Qian, Hangcheng Cao, Qingchuan Zhao. FIGhost: Fluorescent Ink-based Stealthy and Flexible Backdoor Attacks on Physical Traffic Sign Recognition. *IEEE Transactions on Dependable and Secure Computing*, 2026. (**CCF A**)

[4] **[TDSC 2026]** Rui Zhang, Zihan Wang, Tianli Yang, Wenbo Jiang, Rui Zhang, Qingchuan Zhao, Hongwei Li, Yang Liu, **Guowen Xu***. Hidden Tail: Adversarial Attack for Stealthy Resource Consumption against Vision-Language Models. *IEEE Transactions on Dependable and Secure Computing*, 2026. (**CCF A**)

[5] **[TDSC 2026]** Rui Zhang, Yun Shen, Hongwei Li, Wenbo Jiang, Hanxiao Chen, Yuan Zhang, **Guowen Xu***, Yang Zhang. Backdoor Complications: A Comprehensive Analysis and Mitigation of the Unforeseen Consequences of Backdoor Attacks. *IEEE Transactions on Dependable and Secure Computing*, 2026. (**CCF A**)

[6] **[AAAI 2026]** Xinyuan Qian, Haoyong Wang, Hangcheng Cao, Shuai Yuan, Senkang Hu, Qingchuan Zhao, Hongwei Li, **Guowen Xu***. MartDE: A Privacy-Preserving and Cost-Efficient Evaluation Framework for Data Marketplaces. *The Fortieth AAAI Conference on Artificial Intelligence*, 2026. (**CCF A**)

[7] **[AAAI 2026]** Zihan Wang, Rui Zhang, Yu Liu, Wenshu Fan, Wenbo Jiang, Qingchuan Zhao, Hongwei Li, **Guowen Xu***. MPMA: Preference Manipulation Attack Against Model Context Protocol. *The Fortieth AAAI Conference on Artificial Intelligence*, 2026. (**CCF A**, Oral)

[8] **[AAAI 2026]** Zihan Wang, Rui Zhang, Hongwei Li, Wenshu Fan, Wenbo Jiang, Qingchuan Zhao, **Guowen Xu***. ConfGuard: A Simple and Effective Backdoor Detection for Large Language Models. *The Fortieth AAAI Conference on Artificial Intelligence*, 2026. (**CCF A**)

[9] **[ACL 2026]** Rui Zhang, Hongwei Li, Yun Shen, Xinyue Shen, Wenbo Jiang, **Guowen Xu***, Yang Liu, Michael Backes, Yang Zhang. The Art of (Mis)alignment: How Fine-Tuning Methods Effectively Misalign and Realign LLMs in Post-Training. *The 64th Annual Meeting of the Association for Computational Linguistics*, 2026. (**CCF A**)

[10] **[TDSC 2025]** **Guowen Xu**, Shengmin Xu, Jianting Ning, Xinyi Huang, Hongwei Li, Rongxing Lu. New Secure Sparse Inner Product with Applications to Machine Learning. *IEEE Transactions on Dependable and Secure Computing*, 2025. (**CCF A**)

[11] **[TDSC 2025]** Jianfei Sun, **Guowen Xu***, Hongwei Li, Tianwei Zhang, Cong Wu, Xuehuan Yang, Robert H. Deng. Sanitizable Cross-domain Access Control with Policy-driven Dynamic Authorization. *IEEE Transactions on Dependable and Secure Computing*, 2025. (**CCF A**)

[12] **[TDSC 2025]** Jianfei Sun, **Guowen Xu***, Yang Yang, Xuehuan Yang, Cong Wu, Zhen Liu, Guomin Yang, Robert H. Deng. Forward-Secure Hierarchical Delegable Signature for Smart Homes. *IEEE Transactions on Information Forensics and Security*, 2025. (**CCF A**)

[13] **[NeurIPS 2025]** Shuai Yuan, Xingshuo Han, Hongwei Li, **Guowen Xu***, Wenbo Jiang, Tao Ni, Qingchuan Zhao, Yuguang Fang. The Fluorescent Veil: A Stealthy and Effective Physical Adversarial Patch Against Traffic Sign Recognition. *The Thirty-Ninth Annual Conference on Neural Information Processing Systems*, 2025. (**CCF A**)

[14] **[ICML 2025]** Rui Zhang, Yun Shen, Hongwei Li, Wenbo Jiang, Hanxiao Chen, Yuan Zhang, **Guowen Xu***, Yang Zhang. The Ripple Effect: On Unforeseen Complications of Backdoor Attacks. *International Conference on Machine Learning*, 2025. (**CCF A**)

[15] **[ICML 2025]** Shuai Yuan, Hongwei Li, Rui Zhang, Hangcheng Cao, Wenbo Jiang, Tao Ni, Wenshu Fan, Qingchuan Zhao, **Guowen Xu***. Omni-Angle Assault: An Invisible and Powerful Physical Adversarial Attack on Face Recognition. *International Conference on Machine Learning*, 2025. (**CCF A**)

[16] **[TDSC 2025]** Xiaoyuan Liu, Hongwei Li, **Guowen Xu***, Shengmin Xu, Xinyi Huang, Tianwei Zhang, Yijing Lin, Jianying Zhou. Antelope: Fast and Secure Neural Network Inference. *IEEE Transactions on Dependable and Secure Computing*, 2025. (**CCF A**)

[17] **[TDSC 2025]** Shuai Yuan, **Guowen Xu***, Hongwei Li, Rui Zhang, Hangcheng Cao, Xinyuan Qian, Tao Ni, Qingchuan Zhao, Yuguang Fang. No Trespassing: Ground-view Adversarial Patches for Privacy-aware Management in COTS Robot Vacuum Cleaner. *IEEE Transactions on Dependable and Secure Computing*, 2025. (**CCF A**)

[18] **[ASIACCS 2025]** Hangcheng Cao, **Guowen Xu***, Wenbing Huang, Hongwei Li. Can Small-scale Evaluation Reflect Real Ability? A Performance Study of Emerging Biometric Authentication. *Proceedings of ACM ASIACCS*, Ha Noi, Vietnam, 2025.

[19] **[ACSAC 2025]** Yuan Zhang, Yali Wang, **Guowen Xu***, Yaqing Song, Hongwei Li. Belt and Braces! Fight against Key Compromising in Single Sign-On Systems. *Proceedings of Annual Computer Security Applications Conference*, Honolulu, Hawaii, USA, 2025. (**CCF B**)

[20] **[TDSC 2024]** Hao Ren, **Guowen Xu***, Tianwei Zhang, Jianting Ning, Xinyi Huang, Hongwei Li, Rongxing Lu. Efficiency Boosting of Secure Cross-platform Recommender Systems over Sparse Data. *IEEE Transactions on Dependable and Secure Computing*, 2024. (**CCF A**)

[21] **[DSN 2024]** Xiaoxuan Lou, Kangjie Chen, **Guowen Xu***, Han Qiu, Shangwei Guo, Tianwei Zhang. Protecting Confidential Virtual Machines from Hardware Performance Counter Side Channels. *The 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks*, 2024. (**CCF B**)

[22] **[EuroS&P 2024]** Guanlin Li, **Guowen Xu***, Han Qiu, Shangwei Guo, Run Wang, Jiwei Li, Tianwei Zhang, Rongxing Lu. Fingerprinting Image-to-Image Generative Adversarial Networks. *IEEE European Symposium on Security and Privacy*, 2024.

[23] **[TDSC 2023]** **Guowen Xu**, Xingshuo Han, Tianwei Zhang, Shengmin Xu, Jianting Ning, Xinyi Huang, Hongwei Li, Robert H. Deng. SIMC 2.0: Improved Secure ML Inference Against Malicious Clients. *IEEE Transactions on Dependable and Secure Computing*, 2023. (**CCF A**)

[24] **[TDSC 2023]** **Guowen Xu**, Xingshuo Han, Gelei Deng, Tianwei Zhang, Shengmin Xu, Jianting Ning, Anjia Yang, Hongwei Li. VerifyML: Obliviously Checking Model Fairness Resilient to Malicious Model Holder. *IEEE Transactions on Dependable and Secure Computing*, 2023. (**CCF A**)

[25] **[TIFS 2023]** **Guowen Xu**, Shengmin Xu, Jinhua Ma, Jianting Ning, Xinyi Huang. An Adaptively Secure and Efficient Data Sharing System for Dynamic User Groups in Cloud. *IEEE Transactions on Information Forensics and Security*, 2023. (**CCF A**)

[26] **[TIFS 2023]** Jianfei Sun, **Guowen Xu***, Tianwei Zhang, Xuehuan Yang, Mamoun Alazab, Robert H. Deng. Privacy-aware and Security-enhanced Efficient Matchmaking Encryption. *IEEE Transactions on Information Forensics and Security*, 2023. (**CCF A**)

[27] **[ICLR 2023]** Guanlin Li, **Guowen Xu***, Shangwei Guo, Han Qiu, Jiwei Li, Tianwei Zhang. Extracting Robust Models with Uncertain Examples. *International Conference on Learning Representations*, 2023. (**CCF A**)

[28] **[TCSVT 2023]** **Guowen Xu**, Guanlin Li, Shangwei Guo, Tianwei Zhang, Hongwei Li. Secure Decentralized Image Classification with Multiparty Homomorphic Encryption. *IEEE Transactions on Circuits and Systems for Video Technology*, 2023. (**CCF B**)

[29] **[CCS 2022]** Gelei Deng, **Guowen Xu***, Yuan Zhou, Tianwei Zhang, Yang Liu. On the (In) Security of Secure ROS2. *Proceedings of ACM Conference on Computer and Communications Security*, 2022. (**CCF A**)

[30] **[TDSC 2022]** **Guowen Xu**, Xingshuo Han, Shengmin Xu, Tianwei Zhang, Hongwei Li, Xinyi Huang, Robert H. Deng. Hercules: Boosting the Performance of Privacy-preserving Federated Learning. *IEEE Transactions on Dependable and Secure Computing*, 2022. (**CCF A**)

[31] **[TIFS 2022]** Jianfei Sun, **Guowen Xu***, Xuehuan Yang, Tianwei Zhang, Mamoun Alazab, Robert H. Deng. Verifiable, Fair and Privacy-preserving Broadcast Authorization for Flexible Data Sharing in Clouds. *IEEE Transactions on Information Forensics and Security*, 2022. (**CCF A**)

[32] **[TIFS 2022]** Jianfei Sun, **Guowen Xu***, Tianwei Zhang, Mamoun Alazab, Robert H. Deng. A Practical Fog-based Privacy-preserving Online Car-hailing Service System. *IEEE Transactions on Information Forensics and Security*, 2022. (**CCF A**)

[33] **[ECCV 2022]** Guanlin Li, **Guowen Xu***, Han Qiu, Ruan He, Jiwei Li, Tianwei Zhang. Improving Adversarial Robustness of 3D Point Cloud Classification Models. *European Conference on Computer Vision*, 2022. (**CCF B**)

[34] **[TITS 2022]** Jianfei Sun, **Guowen Xu***, Tianwei Zhang, Xiaochun Cheng, Xingshuo Han, MingJian Tang. Secure Data Sharing with Flexible Cross-domain Authorization in Autonomous Vehicle Systems. *IEEE Transactions on Intelligent Transportation Systems*, 2022. (**CCF B**)

[35] **[INFOCOM 2021]** Haoran Yuan, Xiaofeng Chen, **Guowen Xu***, Jianting Ning, Joseph Liu, Robert H. Deng. Efficient and Verifiable Proof of Replication with Fast Fault Localization. *Proceedings of IEEE International Conference on Computer Communications*, 2021. (**CCF A**)

[36] **[TCC 2021]** Jianfei Sun, **Guowen Xu***, Tianwei Zhang, Hu Xiong, Hongwei Li, Robert H. Deng. Share your data carefree: An efficient, scalable and privacy-preserving data sharing service in cloud computing. *IEEE Transactions on Cloud Computing*, 2021.

[37] **[TDSC 2020]** **Guowen Xu**, Hongwei Li, Yun Zhang, Shengmin Xu, Jianting Ning, Robert H. Deng. Privacy-preserving Federated Deep Learning with Irregular Users. *IEEE Transactions on Dependable and Secure Computing*, 2020. (**CCF A**)

[38] **[ASIACCS 2020]** **Guowen Xu**, Hongwei Li, Shengmin Xu, Hao Ren, Kan Yang, Yinghui Zhang, Jianfei Sun, Robert H. Deng. Catch You If You Deceive Me: Verifiable and Privacy-aware Truth Discovery in Crowd Sensing Systems. *Proceedings of ACM ASIA Conference on Computer and Communications Security*, Taipei, Taiwan, China, 2020.

[39] **[ACSAC 2020]** **Guowen Xu**, Hongwei Li, Hao Ren, Jianfei Sun, Shengmin Xu, Jianting Ning, Haomiao Yang, Kan Yang, Robert H. Deng. Secure and Verifiable Inference in Deep Neural Networks. *Annual Computer Security Applications Conference*, 2020. (**CCF B**)

[40] **[TCC 2020]** **Guowen Xu**, Hongwei Li, Hao Ren, Xiaodong Lin, Xuemin (Sherman) Shen. DNA Similarity Search with Access Control over Encrypted Cloud Data. *IEEE Transactions on Cloud Computing*, 2019.

[41] **[ICPADS 2020]** **Guowen Xu**, Hongwei Li, Yuan Zhang, Xiaodong Lin, Robert H. Deng, Xuemin (Sherman) Shen. A Deep Learning Framework Supporting Model Ownership Protection and Traitor Tracing. *IEEE International Conference on Parallel and Distributed Systems*, 2020. (Best Paper Award)

[42] **[TIFS 2019]** **Guowen Xu**, Hongwei Li, Sen Liu, Kan Yang, Xiaodong Lin. VerifyNet: Secure and Verifiable Federated Learning. *IEEE Transactions on Information Forensics and Security*, 2019. (**CCF A**)

[43] **[TVT 2019]** **Guowen Xu**, Hongwei Li, Sen Liu, Mi Wen, Rongxing Lu. Efficient and Privacy-preserving Truth Discovery in Mobile Crowd Sensing Systems. *IEEE Transactions on Vehicular Technology*, 2019.

[44] **[IEEE Commun Mag 2019]** **Guowen Xu**, Hongwei Li, Hao Ren, Kan Yang, Robert H. Deng. Data Privacy and Security in Deep Learning: Attacks, Solutions and Opportunities. *IEEE Communications Magazine*, 2019.

[45] **[TIFS 2018]** **Guowen Xu**, Hongwei Li, Yuanshun Dai, Kan Yang, Xiaodong Lin. Enabling Efficient and Geometric Range Query with Access Control over Encrypted Spatial Data. *IEEE Transactions on Information Forensics and Security*, 2018. (**CCF A**)

[46] **[CCS 2018]** **Guowen Xu**, Hongwei Li, Rongxing Lu. Poster: Practical and Privacy-Aware Truth Discovery in Mobile Crowd Sensing Systems. *Proceedings of ACM Conference on Computer and Communications Security*, Toronto, Canada, 2018. (**CCF A**)

[47] **[Comput Secur 2017]** **Guowen Xu**, Hongwei Li, Chen Tan, Dongxiao Liu, Yuanshun Dai, Kan Yang. Achieving Efficient and Privacy-Preserving Truth Discovery in Crowd Sensing Systems. *Computers & Security*, 2017. (**CCF B**)

PROJECT EXPERIENCES

[1] **[2027.01–2029.12]** Research on Key Technologies of Full-Link Security Evaluation and Protection for Intelligent Agents, Young Teachers Fund of the Fok Ying Tung Education Foundation, PI, Amount: RMB 180K.

[2] **[2026.06–2027.06]** Research on Adversarial Defense for Multimodal Large Language Models Based on Dynamic Coherent Ising Machines, CAAI–QBoson Quantum Computing Application Innovation Fund, PI, Amount: RMB 200K

[3] **[2026.01–2028.12]** Research on Security-Critical Technologies for Autonomous Driving Based on Multimodal Large Models, National Science Fund Program for Young Scientists of China, PI, Amount: RMB 300K.

[4] **[2026.01–2029.12]** XXX Platform Design and Implementation. Key R&D Project of XXX. PI, Amount: RMB 9.6 million.

[5] **[2026.01–2027.12]** Research on Key Technologies for Integrated Attack-Defense of Multimodal Large Models for Autonomous Driving. General Program of the Sichuan Provincial Natural Science Foundation. PI, Amount: RMB 150K.

[6] **[2025.01–2027.12]** Data Security, National Science Fund Program for Excellent Young Scientists (Overseas) of China, PI, Amount: RMB 2 million.

[7] **[2024.09–2027.09]** Key Security and Privacy Issues on Machine Learning, PI, Start-up at UESTC, Amount: RMB 2 million.

ACADEMIC ACTIVITIES

Editorial Board:

- IEEE Transactions on Multimedia, 2026–present
- IEEE Transactions on Pattern Analysis and Machine Intelligence, 2025–present
- IEEE Transactions on Dependable and Secure Computing, 2025–present
- IEEE Transactions on Information Forensics and Security, 2024–present
- ACM Computing Surveys, 2026–present
- IEEE Transactions on Consumer Electronics, 2026–present
- IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2025–present
- IEEE Transactions on Circuits and Systems for Video Technology, 2024–present
- IEEE Transactions on Network and Service Management, 2023–2026
- IEEE Open Journal of Signal Processing, 2023–2026
- ACM Digital Threats: Research and Practice, 2023–2026
- Pattern Recognition, Elsevier, 2024–present
- Information Fusion, Elsevier, 2024–present
- Cybersecurity, Springer, 2024–present
- Lead Guest Editor of ACM Transactions on Autonomous and Adaptive Systems, Special Issue on Trustworthy Security and Privacy-AI Powered Autonomous Driving, 2024–2025
- Lead Guest Editor of IEEE Transactions on Industry Applications, Special Issue on Advancing Intelligent Control, Resilience, and Flexibility in Modern Industrial and Power Systems (ACRF), 2026–2028

Technical Committee:

- 2027 Area Chair of International Conference on Machine Learning
- 2027 Area Chair of AAAI Conference Artificial Intelligence
- 2027 Area Chair of International Joint Conference on Neural Networks
- 2027 Program Committee Member of Network and Distributed System Security Symposium
- 2027 Program Committee Member of IEEE European Symposium on Security and Privacy
- 2026 Area Chair of Neural Information Processing Systems
- 2026–2028 Member of IEEE Computer Society Standards Activities Board Outreach & Communications Standing Committee
- 2026–2028 Member of IEEE Region 10 Young Professionals Committee
- 2026 Associate Chair ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing
- 2026 Associate Chair of ACM CHI Conference on Human Factors in Computing Systems
- 2026 Area Chair of International Conference on Machine Learning
- 2026 Area Chair of IEEE International Conference on Acoustics, Speech, and Signal Processing
- 2026 Area Chair of International Conference on Learning Representations
- 2026 Area Chair of ACM Conference on Knowledge Discovery and Data Mining

- 2026 Senior Program Committee of AAAI Conference Artificial Intelligence
- 2026 Area Chair of International Joint Conference on Neural Networks
- 2026 Program Committee Member of The ACM Conference on Computer and Communications Security
- 2026 Program Committee Member of The 35th USENIX Security Symposium
- 2025 Area Chair of ACM Conference on Knowledge Discovery and Data Mining
- 2025 Area Chair of International Conference on Machine Learning
- 2025 Associate Chair of ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing
- 2025 Area Chair of International Conference on Learning Representations
- 2025 Senior Program Committee of AAAI Conference Artificial Intelligence
- 2025 Area Chair of International Joint Conference on Neural Networks
- 2025 The 35th IEEE Visualization and Visual Analytics Conference
- 2025 Program Committee Member of The 37th International Conference on Computer Aided Verification
- 2025 Program Committee Member of The 32nd IEEE Conference on Virtual Reality
- 2025 International Conference on Autonomous Agents & Multiagent Systems
- 2025 Program Committee Member of The 30th Annual ACM Conference on Intelligent User Interfaces
- 2025 Program Committee Member of The 29th Financial Cryptography and Data Security Conference
- 2025 Program Committee Member of IEEE European Symposium on Security and Privacy
- 2024 Area Chair of International Conference on Machine Learning
- 2024 Program Committee Member of Thirty-eighth Conference on Neural Information Processing Systems
- 2024 Program Committee Member of Annual Computer Security Applications Conference
- 2023 Program Committee Member of The ACM Web Conference
- 2023 Program Committee Member of Thirty-Seventh AAAI Conference on Artificial Intelligence
- 2022 Program Committee Member of Thirty-sixth Conference on Neural Information Processing Systems

Others:

- Technical Review Board of IEEE Transactions on Parallel and Distributed Systems, 2025–present
- Distinguished Reviewers Board of ACM Transactions on the Web, 2022–present
- Mentor of IEEE Signal Processing Mentorship Academy, 2025–present